Sales Data Package

1. Overview

This SSIS package handles the ETL (Extract, Transform, Load) process for importing raw data from a flat file into a SQL Server database. The workflow includes data type conversion, sorting, business logic application via derived columns, and a centralized error-handling mechanism.

Sales.csv

transaction_id,customer_id,product_category,item_name,quantity,unit_price,sale_date,store_location,payment_method

TXN_1000,138,Electronics,Desk Lamp,4,779.88,2024-03-12, Miami ,Cash

TXN_1001,128,Books,Hardcover Novel,1,224.88,2024-03-10,Houston,Credit Card

TXN_1002,114,Electronics,Laptop,2,1129.44,2024-03-10,New York,

TXN_1003,142,Books,Wall Art,1,1145.41,2024-03-05, Miami ,Credit Card

TXN_1004,107,Books,Laptop,4,1099.11,2024-03-07,New York,

TXN_1005,120,Books,Laptop,2,453.64,2024-03-04,Chicago,Credit Card

TXN_1010,110,Books,Hardcover Novel,-5,1156.89,2024-03-07,New York,PayPal

TXN_1020,129,Books,Cotton T-Shirt,4,,2024-03-03,Houston,Credit Card

TXN_1030,148,Apparel,Smartphone,2,850.00,2024/13/45,Los Angeles,PayPal

TXN_1000,138,Electronics,Desk Lamp,4,779.88,2024-03-12, Miami ,Cash

TXN_1001,128,Books,Hardcover Novel,1,224.88,2024-03-10,Houston,Credit Card

TXN_1031,105,Home Decor,Wall Art,3,45.50,2024-03-08,Chicago,Cash

TXN_1032,112,Electronics,Smartphone,1,999.99,2024-03-09, Miami ,PayPal

TXN_1033,140,Apparel,Cotton T-Shirt,2,25.00,2024-03-11,Houston,Credit Card

2. Data Flow Architecture

The Data Flow Task consists of the following components:

A. Data Extraction & Transformation

- **Flat File Source**

Description: Extracts raw data from a source file (e.g., CSV or TXT).

Current Execution: Processed 14 rows.

- **Data Conversion:**

Description: Converts data types from the source (usually string) to the required types for the destination database.

Output: 13 rows successfully converted; 1 row redirected to error handling.

- **Sort:**

Description: Reorders the dataset based on specific columns. This may be used to handle duplicates or prepare data for efficient loading.

Input: 13 rows.

- **Derived Column:**

Description: Applies transformation logic, such as string manipulation, calculations, or adding metadata (e.g., load timestamps).

Output: 11 rows passed to the destination.(duplicates removed during sort)

B. Data Loading

- **OLE DB Destination:**

Description: Maps the transformed data to the target SQL Server table.

Current Execution: Successfully loaded the processed dataset.

C. Error Handling Strategy

The package utilizes a "Redirect Row" error handling configuration:

Union All (Error Collector): Collects failed rows from both the Data Conversion and OLE DB Destination components into a single data stream.

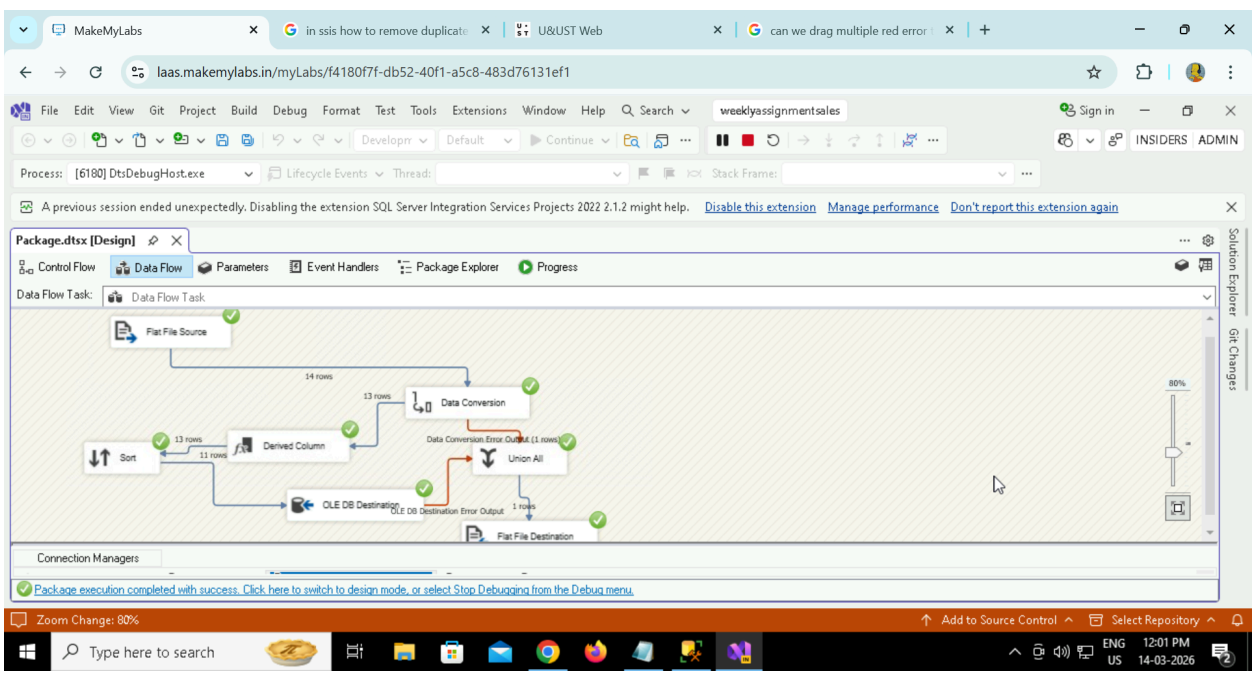
Flat File Destination (Error Log): All failed records (including error codes and column IDs) are written to a secondary file for manual review and troubleshooting.

Audit Summary: In the current run, 1 record failed during the conversion stage and was logged.

3. Connectivity

- **Connection Managers:**

Flat File Connection Manager: Points to the source data file.

OLE DB Connection Manager: Points to the target SQL Server Instance and Database.

Error Log Connection: Points to the CSV file used to store rejected records.